

Geopolitical conflict and risk and the EU energy trading: A dynamic evolutionary networks analysis[☆]

Shuanglei Xu^a, Youyi Deng^a, Rabindra Nepal^b, Tooraj Jamasb^{c,*}

^a School of International Trade and Economics, University of International Business and Economics, Beijing 100029, China

^b School of Business, Faculty of Arts, Society and Business, University of Wollongong, New South Wales, Australia

^c Copenhagen School of Energy Infrastructure, Department of Economics, Copenhagen Business School, Denmark

ARTICLE INFO

JEL classifications:

F18

Q43

Keywords:

European Union

Conventional energy trade

Renewable energy trade

Social network

Exponential random graph model

ABSTRACT

The European Union's (EU) energy imports have continued to face challenges while heightening since the Russia-Ukraine conflict, and therefore energy security has come to the fore. In this study, we adopt a social network approach (SNA) and an exponential random graph model (ERGM) to analyze the impact of the conflict on the energy trade focusing on the EU and its relations with major energy trading countries. We use data from a sample of 47 countries from 2014 to 2023 to explore the characteristics of the structural evolution of the EU's conventional and renewable trade networks and the influencing mechanisms behind them. The EU's conventional network is undergoing a contraction as the conflicts are driving a significant global trend towards decoupling. Meanwhile, its renewable trade network is thriving, indicating a shift in energy structures and the restructuring of the core trade periphery. Russia is fading out of the core circle of trade, and the US is becoming a key hub connecting all parties. Germany, France, and the Netherlands play the role of essential importers as core nodes of the network. Mechanistic analysis shows that mutualism plays a vital role in multilateral trade. Rising geopolitical risks, while posing a barrier to energy imports, have facilitated a boom in renewable trade. Economic size and trade openness have positively driven energy trade. Foreign investment, intellectual property rights, and population and urbanization levels have differentiated the two types of energy trade. Geographic proximity, linguistic commonality, and free trade agreements positively contributed to the construction and maintenance of energy trade networks. This study depicts the dynamics of EU energy trade under geopolitical turbulence and informs the transformation of the EU's energy structure.

1. Introduction

The stable supply of oil and natural gas is a cornerstone of national strategic resources and is directly related to national security, social stability, and economic development, while energy security is increasingly becoming a global issue (Khurshid et al., 2024; Jiao et al., 2023; Acheampong et al., 2021). Energy security encompasses several dimensions, and EU policies, in particular, emphasize the centrality of the security of the energy supply (Labelle, 2024; Wei et al., 2022). Trade can regulate energy supply and demand implying that a country can import energy from other regions when the country's supply is insufficient. As an extreme geopolitical conflict, the Russia-Ukraine conflict has pushed up energy prices and triggered a global energy crisis (Ma et al., 2023; Jiao et al., 2023; Wu et al., 2023). The vulnerability of the EU's energy

security is exacerbated by the fact that its conventional energy supply is not sufficient and that it is highly dependent on imports; in particular, 60% of its energy imports are from Russia (Chen et al., 2023; Desogus et al., 2023).

Energy security is linked to geopolitical relations. Geopolitical risks stemming from war, terrorism, and inter-state tensions constitute potential threats to politics, economies, militaries, and societies, and globalization has further amplified these risks (Caldara and Iacoviello, 2022; Jawadi et al., 2024). In response to this challenge, the EU is actively pursuing an energy transition that involves reducing the use of "high-carbon" energy sources and promoting the use of "low-carbon" renewable energy sources (Blondeel et al., 2021; Zhang et al., 2023). On the one hand, the EU is committed to diversifying its energy supply by looking for alternative sources to Russian natural gas, such as increased

[☆] This article is part of a Special issue entitled: 'Geopolitical Conflicts' published in Energy Economics.

* Corresponding author.

E-mail addresses: xushuanglei1023@126.com (S. Xu), 18838956300@163.com (Y. Deng), rnepal@uow.edu.au (R. Nepal), tj.eco@cbs.dk (T. Jamasb).

imports of US liquefied natural gas. Although this initiative has eased the short-term supply pressure, it has brought new problems of rising costs and impeded energy mix transition (Chen et al., 2023). On the other hand; geopolitical risks have prompted the EU to undertake a green energy transition; aiming at long-term green and low-carbon development and reducing external dependence (Cai and Wu, 2021; Khan et al., 2023a; Cai et al., 2022; Panarello and Gatto, 2023). Solar photovoltaic is the EU's vigorous development of renewable energy. China impacts the reconstruction of the EU energy system as the largest exporter of solar photovoltaic panels, (Yan et al., 2023).

A combination of internal and external factors has shaped the reconfiguration of the EU energy system. Internally, the EU is gradually decreasing its dependence on imports of conventional energy sources in favor of increased imports of new energy sources (Carodenuto et al., 2024) through increasing environmental regulations (Jang and Yang, 2024); the removal of fossil fuel subsidies (Antimiani et al., 2023); increasing subsidies for renewable energy (Moiseyev et al., 2014); and strict enforcement of trade regulations and widespread signing of trade agreements (Apeti and N'doua, 2023; Zhou et al., 2019; Carodenuto et al., 2024). Member states' energy preferences influence trade flows, mainly as increased awareness of climate change drives more countries to favor solar and wind energy (Arndt, 2023; Batzella, 2024). Among the external factors, geopolitical risks such as regional conflicts and trade sanctions challenge the global energy supply chain and network (Ren et al., 2024). These risks have impacted energy trade and security by affecting the supply-demand balance in importing and exporting countries; leading to dramatic market fluctuations (Liu et al., 2021b). In particular; they have had a significant impact on conventional energy markets; pushing up prices and reducing trade volumes and resource control (Liu et al., 2021a; Qian et al., 2022; Nakhli et al., 2021; Dong and Li, 2018; Wang et al., 2023; Zhang et al., 2021). Geopolitically sensitive regions such as the Middle East have been forced into the energy game, which has even reshaped the energy trade landscape (Zhang et al., 2024b; Gong et al., 2023; Estrada and Koutronas, 2022; Zheng et al., 2022). The sanctions imposed by the US and the EU on Russian energy in the aftermath of the Russia-Ukraine conflict (Chen et al., 2023; Qureshi et al., 2022) have triggered energy shortages, threatening the energy security of the EU (Zheng et al., 2022; Ngo et al., 2024; Labelle, 2024; Owjimehr et al., 2023). As a result, the EU is forced to accelerate decoupling from Russian energy and to accelerate energy transition.

Geopolitical risk has a dual effect on the reconfiguration of the energy system. It exacerbates the instability of conventional energy markets and provides offers for developing renewable energy (Su et al., 2021; Lee et al., 2024). There are differences between renewable and fossil energy at the geopolitical level (Maghyereh et al., 2019). In the future; technology; rather than conventional resources; will be the central force dominating the reconfiguration of the energy system (Bazilian et al., 2020). Geopolitical tensions have further prompted significant powers to emphasize the strategic position of new energy (Gao et al., 2015; Scholten et al., 2020) and to use international trade as a key factor of renewable energy development (Khan et al., 2017; Alam and Murad, 2020). In response to geopolitical risks, large energy-consuming countries are restructuring their energy mix to increase the use of renewable energy (Desogus et al., 2023; Vakulchuk et al., 2020; Owjimehr et al., 2023). Developed countries, in particular, prefer to mitigate the threat of energy shortages by importing renewable energy sources (Sweidan, 2021). This transition is accompanied by high infrastructure and technology investment costs; which may cause socioeconomic volatility in the short term (Nguyen et al., 2024). In the long run; this transition is crucial for building a sustainable and environmentally friendly energy system (Kivimaa and Sivonen, 2023); providing an essential pathway for the EU to address geopolitical challenges (Khan et al., 2023b; Filipović et al., 2022; Panarello and Gatto, 2023).

Existing research has extensively explored how geopolitics shapes the landscape of conventional energy trade while shedding initial light on the unique impact of geopolitics in the renewable energy sector.

However, most of these explorations have been limited to the unilateral effects of conventional energy markets and have rarely touched on the dual mechanism of geopolitical risks in renewable energy trade, especially in the energy transition process. In addition, most studies tend to favor a global perspective, ignoring the variability of interregional trade patterns. The use of econometric regression methods mainly in existing studies may also underestimate the actual impacts because they must capture complex trade relations between countries. Critically, there is a gap in exploring the dynamic influences of geopolitical risks on the EU's trade in conventional and renewable energy over the long term. This study innovatively focuses on the EU and its complex trade relations with major energy countries, adopting SNA and ERGM as analytical tools and a sample of 47 countries from 2014 to 2023, aiming to reveal the characteristics of the structural evolution of the EU's conventional and renewable energy trade networks and the impact mechanisms behind them.

Our study is innovative in three major ways. First, this study subtly shifts to a regional perspective, focusing on the EU and providing insights into how its trade networks have evolved in the complex context of geopolitics in contrast to the widely adopted macro-view of global trade networks. Hence, this shift dramatically enhances our understanding of EU trade dynamics and geopolitical relevance.

Second, traditional studies focus on direct factors such as economic factors and geographical distance. This study takes a different approach by delving into how geopolitical factors act as key factors that shape and drive changes in the structure of the EU's conventional and renewable energy trade networks. This shift in perspective informs the prognosis of the future direction of EU energy trade.

The third innovation originates from the methodological approach. This study breaks through the current quantitative research dominated by econometric methods such as fixed-effects models and breakpoint regressions and innovatively introduces SNA, combined with ERGM (Exponential Random Graph Model) and TERGM (Temporal Exponential Random Graph Model), to realize a multidimensional exploration of the evolution of the EU energy trade network. This methodological innovation not only enriches the research tools in the field of energy trade but also opens up a new path for upgrading the EU's energy structure and the development of energy trade from the dual perspective of static and dynamic evolution.

The reminder of this study is structured as follows while Fig. 1 below shows the framework of this paper. Section 2 presents the research hypothesis. Section 3 uses the SNA to analyze the trade network at the overall and individual levels. Section 4 constructs exponential random graph models to examine internal and external factors of trade network evolution and conducts model validity and robustness tests. Section 5 concludes the paper while offering policy recommendations.

2. Research hypothesis

This paper further applies ERGM and TERGM to analyze the mechanism of trade network evolution from both static and dynamic dimensions based on the EU conventional and renewable energy trade network. The model's core independent variables include three network structural characteristics: endogenous structural characteristics, node attribute effects, and external environment effects (Lu and Li, 2023). The sub-sections below individually explores the mechanism of their influence on trade networks from these three dimensions.

2.1. Endogenous structure

SNA reveals the interdependence within a network by highlighting how pre-existing network relationships shape the formation and development of new relationships as commonly known as the endogenous structure effect (Zhang et al., 2024a). This study selected edges and mutual as key endogenous variables and included them in the analysis (Leifeld et al., 2018). The edges are like a constant term in linear

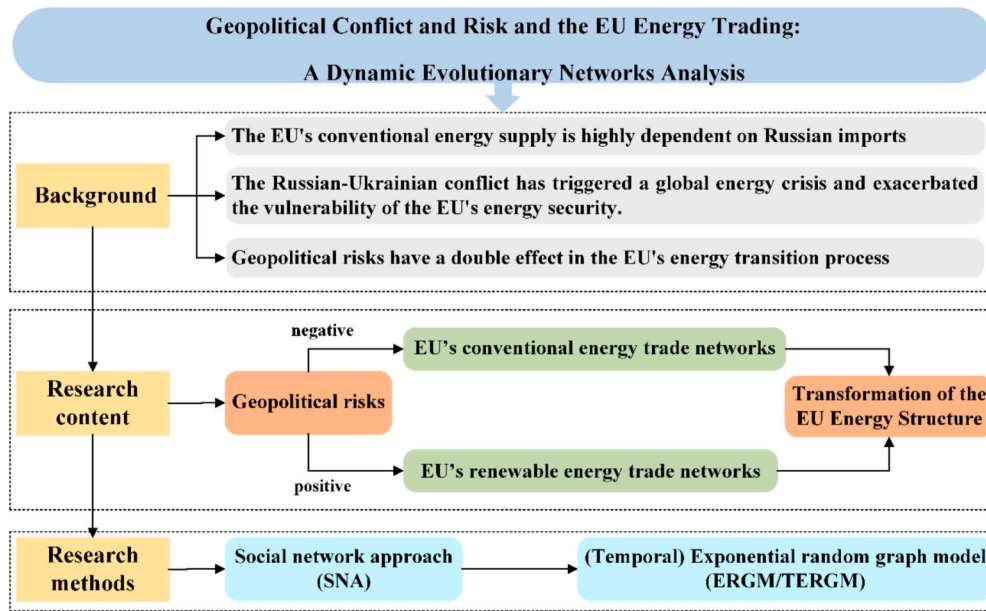


Fig. 1. Theoretical framework.

regression and represent the baseline likelihood of network formation. Reciprocity measures the closeness of bilateral trade; and networks with high reciprocity tend to be more robust; reflecting the frequency and depth of trade between economies (Zhang et al., 2024a). No assumptions are required given that these two variables are inherent attributes of the network.

2.2. Node attributes

Geopolitical risk covers factors such as war, terrorism, and other factors jeopardizing international peace and stability (Gao and Xu, 2024). It is especially pronounced in energy resource-rich regions; affecting the stability of global energy markets and economic development (Caldara and Iacoviello, 2022). Geopolitical tensions not only have a direct impact on energy security by disrupting the supply and demand systems of conventional energy (Gong et al., 2022) but also incentivize countries to promote the development of new energy sources to reduce dependence on conventional energy (Su et al., 2021). The Russia-Ukraine conflict, for example, has led to the blockage of the EU's conventional energy imports from Russia, which has stimulated the diversification of the EU's conventional energy imports and increased its investment in renewable energy to ensure energy security and facilitated a profound transformation of the energy mix.

In addition, nodal attributes such as economic size, trade openness, foreign direct investment, intellectual property rights, population, and degree of urbanization are critical. GDP is a key external variable that shapes the structure of the trade network as a measure of a country's economic strength (Jang and Yang, 2022); and its growth lays a solid foundation (Guo et al., 2023a). Increased trade openness; in turn; contributes to energy trade (Feng et al., 2021; Guo et al., 2023b). Foreign investment tends to promote industrialization, which leads to increased demand for conventional energy and can provide financial and technical support for renewable energy (Wang et al., 2021).

Increased intellectual property rights offer a protection mechanism for energy technology innovation. Royalties for intellectual property rights facilitate the innovation of energy technologies, which helps drive energy trade in a more efficient and sustainable direction (Dussaix et al., 2022). Population growth triggers demand for conventional energy; thus driving trade growth in conventional energy. However; countries increase investment and promote renewable energy; thereby reducing imports to meet the growing energy demand in the long run

(Voumik et al., 2023). Higher urbanized countries with dense populations and high economic activity have stronger energy demand; thus driving the growth of trade in conventional energy. However; since renewables are not as stable as conventional ones; this may negatively impact renewable energy imports (Asghar et al., 2024; Li and Hane-klaus, 2022).

H1: EU imports conventional energy while boosting renewable energy trade. Node attributes such as scale, trade openness, and foreign direct investment influence the evolution of energy trade networks.

2.3. External networks

The evolution of trade networks is influenced by environmental factors such as geographic proximity, language commonality, and free trade agreements (Li et al., 2024). The geographic proximity helps reduce transportation costs and facilitate trade. For example; Russia's proximity to the EU makes the EU highly dependent on Russia for conventional energy imports (Zhang et al., 2024a). Language; as a bridge of cultural exchange; can directly affect the convenience and efficiency of trade (Zhang et al., 2024a). A common language facilitates smooth communication between countries; enhancing the vitality of bilateral trade (Feng et al., 2021; Zhang et al., 2024a). Countries conclude free trade agreements (FTA) and construct an extensive FTA network (Wu et al., 2020) to reduce communication barriers and trade costs and further promote the prosperity of bilateral trade (Xu et al., 2021).

H2: Geographic proximity, language commonality, and free trade agreement networks positively impact the establishment and maintenance of energy trade networks.

3. Network construction and evolution analysis

Existing literature predominantly focuses on conventional energy research and rarely analyzes energy trade among countries like the EU from a regional perspective or using network methodologies. Therefore, this study first employs social network analysis (see section 3) to construct trade networks for both conventional and renewable energy within the EU. This approach delineates the structural characteristics of energy trade between member states. Simultaneously, it establishes a quantitative foundation (see section 4), which utilizes exponential random graph models (static and dynamic) to analyze network

evolution mechanisms and their influencing factors.

3.1. Network construction

The bilateral trade model is insufficient to fully reveal the complex trade interactions between countries, so scholars have turned to the emerging perspective of SNA to explore inter-country trade relations in depth (Zhu et al., 2024). SNA is used to study the relational structure of social networks and their properties; and its application to the field of trade can reveal dynamic features; critical paths; and trade blocs in the structure of trade networks (Yang et al., 2024). Each node in the network represents an individual country; while the edges connecting these nodes symbolize the trade links between them (Zhang et al., 2024a; Zheng et al., 2022; Zhang and Wen, 2024). Countries form an extensive trade network through trade transactions, defined as $G = (V, E, W)$. This paper constructs a directed weighted trade network by focusing on the EU's trade of conventional energy sources (coal, oil, gas) and renewable energy sources (especially wind and photovoltaic). In this model, nodes V refers to the countries that participated in energy trade, edges E refer to the energy trade relations between these countries, and the weights of the edges W reflect the size of the energy trade volume.

3.2. Analysis of the evolution of trade network structure

3.2.1. Overall network analysis

Using the Ucinet software, we analyze several characteristics of trade networks in detail, such as nodes, number of connected edges, network density, aggregation, and reciprocity. Network density, as a key parameter measuring the ratio of actual to potential links in a network, intuitively reflects the overall tightness of the network. Higher density values imply closer trade relations among countries (Cai et al., 2023). The clustering coefficient reveals the proximity between a node and its neighboring nodes; with higher coefficient values reflecting the shortening distance between nodes and enhancing the clustering effect (Zhang et al., 2024a).

Table 1 presents the primary structural characteristics of the EU's conventional and renewable energy trade networks from 2014 to 2023. Forty-seven countries serve as nodes in the conventional EU energy trade network. In 2019, the number of contiguous edges is 507, the density is 0.235, and the reciprocity is 0.313. Due to the epidemic, in 2020, the number of contiguous edges decreased to 481, the density to 0.222, and reciprocity to 0.311. This may stem from disruptions in the supply chain due to embargo measures by various countries, affecting the transportation of energy and cross-border flows, while the global slowdown curbs the energy demand. After 2021, the number of contiguous edges rebounds to 516 and 557 as the economy recovers and demand picks up, and density rises in tandem, suggesting a rapid recovery of energy trade networks in the post-blockade era.

However, the number of network edges in 2023 significantly decreases to 493, the network density slips to 0.228, the clustering coefficient falls to 0.465, and the reciprocity coefficient shrinks to 0.229 as

compared to 2021 due to the Russia-Ukraine conflict in 2022 and the intensifying trend of global decoupling. This data change series points to the weakening of traditional energy trade links. This is because, after the Russia-Ukraine conflict, countries adjusted their energy security strategies and reduced their energy dependence on Russia, which drove the reorganization of the supply chain and led to dynamic changes in the trade network.

In the renewable energy trade network, 47 countries act similarly as nodes. The number of connecting edges is stable in 2019–2020, but aggregation declines to 0.406, reflecting that the epidemic has hit part of the renewable energy supply chain. The network expanded significantly after 2022, with the number of connecting edges reaching 481, the density rising to 0.222, aggregation rebounding to 0.561, and remains high in 2023, suggesting a continued strengthening of EU renewable energy trade links.

Multiple factors drive this trend. In the aftermath of the epidemic, the EU has accelerated its energy transition and tightened its carbon-neutral policies while boosting renewable energy imports. The Russia-Ukraine conflict has further exposed the instability of traditional energy supply chains, prompting countries to increase renewable energy cooperation to reduce energy security risks.

As shown in Figs. 2 and 3, this study has mapped comparative diagrams of conventional and renewable energy trade networks from 2021 to 2023 to visually illustrate the impact of the Russia-Ukraine conflict on the structural characteristics of the EU's energy trade network. The nodes' size intuitively reflects each country's centrality in the trade network implying that the larger the node, the stronger the country's central role in the network. The thickness of the connecting edges directly reflects the trade volume, with thicker lines representing larger volumes of trade with trading partners.

As shown in Fig. 2, the EU's conventional energy trade network contracted overall from 2021 to 2023. The number of connected edges decreased from 516 in 2021 to 493 in 2023, while reciprocity fell from 0.326 to 0.229, reflecting a weakening trend in trade ties. Notably, Russia's node size continued to shrink during this period, with its degree of centrality declining from 41 to 27. This indicates a significant diminishment of Russia's position within the traditional energy trade network, reflecting the pronounced impact of the Russia-Ukraine conflict and global “decoupling” trends on the EU's conventional energy trade structure.

Fig. 3 reveals that the EU renewable energy trade network exhibited pronounced expansion from 2021 to 2023. The number of edges surged from 139 to 454, network density increased from 0.064 to 0.210, and the clustering coefficient rose from 0.473 to 0.553, reflecting strengthened trade ties. During the same period, the degree of centrality of China, the United States, and major EU countries increased, with node sizes significantly expanding. This indicates their continuously strengthened central position within the renewable energy trade network, reflecting the ongoing deepening of renewable energy trade ties driven by both energy transition policies and geopolitical factors.

Overall, the EU's energy trade network exhibited pronounced

Table 1
Descriptive statistics of conventional and renewable energy trade networks in the EU.

Conventional energy trading networks						Renewable Energy Trade Network				
Year	Node	Contiguous	Density	Aggregation	Reciprocity	Node	Contiguous	Density	Aggregation	Reciprocity
2014	47	496	0.229	0.453	0.326	47	131	0.061	0.490	0.323
2015	47	499	0.231	0.456	0.303	47	129	0.060	0.567	0.344
2016	47	496	0.229	0.454	0.278	47	130	0.060	0.506	0.327
2017	47	508	0.235	0.465	0.296	47	122	0.056	0.584	0.312
2018	47	507	0.235	0.469	0.307	47	142	0.066	0.514	0.340
2019	47	507	0.235	0.454	0.313	47	124	0.057	0.486	0.363
2020	47	481	0.222	0.469	0.311	47	126	0.058	0.406	0.355
2021	47	516	0.239	0.487	0.326	47	139	0.064	0.473	0.350
2022	47	557	0.258	0.497	0.323	47	481	0.222	0.561	0.329
2023	47	493	0.228	0.465	0.229	47	454	0.210	0.553	0.331

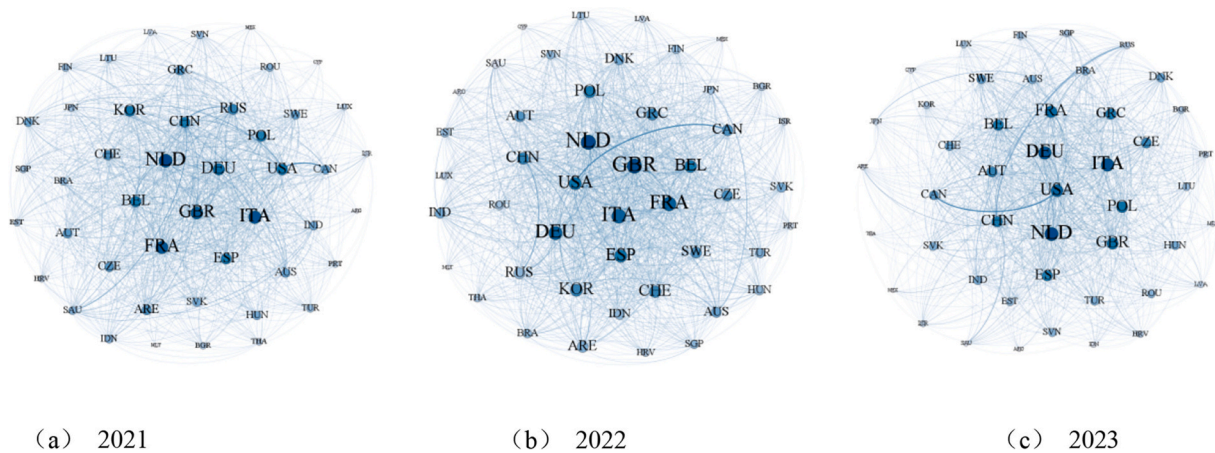


Fig. 2. EU conventional energy trade networks.

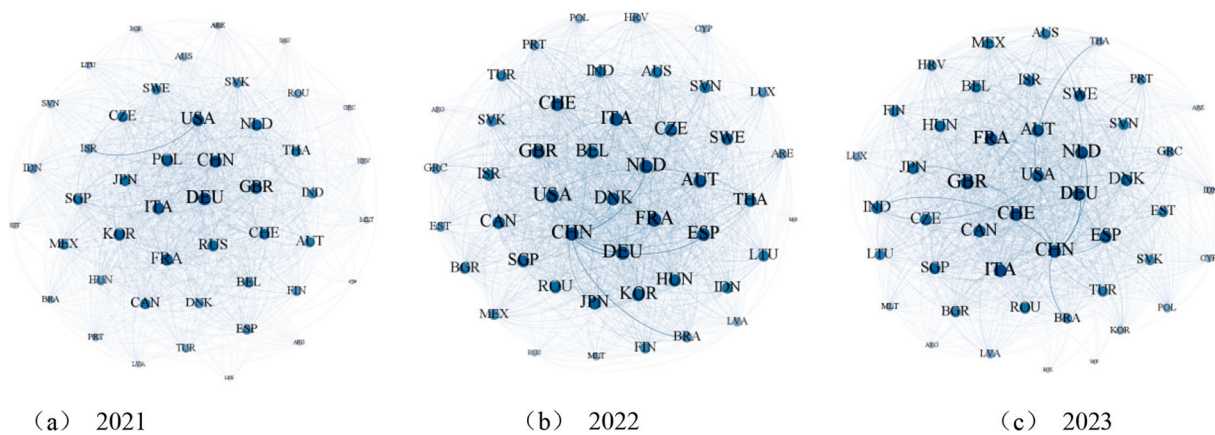


Fig. 3. EU renewable energy trade networks.

structural divergence from 2021 to 2023. The number of connected edges, reciprocity, and core node degree in the conventional energy trade network declined significantly, while the number of connected edges, density, and core nodes in the renewable energy trade network increased markedly. The network structure shifted towards polycentricity, indicating that the Russia-Ukraine conflict accelerated the restructuring of the EU's energy trade framework from conventional to renewable energy sources.

3.2.2. Centrality

The centrality metric measures the extent to which a node acts as a core hub in a network, revealing the node has control over the network. This indicator is usually assessed by combining three dimensions: degree centrality, mediated centrality, and proximity centrality (Freeman, 1979). Economies with a high degree of centrality have extensive trade partnerships and can significantly influence the entire network (Xi et al., 2019). In particular, intermediary centrality emphasizes a country's ability to facilitate trade flows to other countries (Cai et al., 2023). Higher intermediary centrality implies that the country plays a more significant role as a bridge in international trade and bears an amount of trade-connecting information. Given Russia's small energy imports, we focus on the evolution of its export trade relations. For Russia, the US, and China, out-degree (i.e., the number of exporting countries) is a centrality measure. In contrast, the entry degree (i.e., the number of importing countries) is analyzed for the other major EU countries.

The EU's energy trade network has undergone a profound structural adjustment because of the Russia-Ukraine conflict and the global trend towards decoupling, notably characterized by the shrinking of

conventional energy trade links and the dynamic rise of renewable energy trade. Table 2 shows the changes in centrality indicators of the EU's conventional energy trade network from 2021 to 2023. Russia's export influence in the conventional energy market has been hit hard, with its out-degree dropping sharply from 47 in 2021 to 27 in 2023, which is followed by an across-the-board decline in other centrality indicators, showing a significant weakening of Russia's global control over conventional energy supplies, especially in the EU market.

On a country-by-country basis, major conventional energy-importing countries in the EU, such as Germany, France, the Netherlands, the UK, and Italy, while still occupying the center of the network, have accelerated the pace of energy system restructuring in the face of drastic changes in the energy supply landscape. Germany, a highly industrialized country, is heavily dependent on external dependence on three primary non-renewable energy sources, especially the high reliance on Russian energy, which has suffered an unprecedented shock after the Russia-Ukraine conflict. Data show that Germany's imports of natural gas, coal, and oil from Russia accounted for 55%, 50%, and 35% of the market in 2022, but these shares all plummeted to zero by 2023 (Xin and Zhang, 2023). Renewable energy is gradually replacing conventional energy with Germany's rapid development and efficiency gains in renewable energy; especially photovoltaic (PV) technology. The situation is different in France; which has a high share of nuclear energy and is more self-sufficient. But even so, France also has a dependence on conventional energy. The Netherlands has an equally significant dependence on fossil fuels as a powerhouse of energy-intensive industries such as the chemical industry (Astrov et al., 2022).

Table 3 shows the changes in centrality indicators for the EU's

Table 2
Degree of centrality of conventional energy trade networks in the EU.

2021				2022				2023			
country	degree	Closeness	Betweenness	country	degree	Closeness	Betweenness	country	degree	Closeness	Betweenness
EU Conventional Energy Trade Network											
Russia	41	1.26	39.84	Russia	38	0.91	0	Russian	27	0.68	0
United States	37	1.27	170.35	United States	37	1.24	185.94	United States	36	1.07	191.08
China	20	1.14	97.81	China	22	1.10	91.66	China	19	0.97	129.07
Italy	28	1.22	244.28	Italy	27	1.18	176.32	Italy	28	1.13	249.70
Netherlands	19	1.23	146.94	Spain	25	1.19	168.20	Netherlands	28	1.13	152.94
Poland	16	1.10	143.01	Germany	27	1.20	141.91	Spain	21	1.04	118.46
Germany	22	1.22	131.97	Belgium	18	1.14	128.0	Germany	26	1.10	104.22
United Kingdom	17	1.23	131.69	United Kingdom	18	1.15	112.94	United Kingdom	19	1.05	102.60
France	20	1.20	106.08	Netherlands	24	1.16	111.34	Belgium	17	1.00	93.21
Spain	21	1.11	94.11	Greece	15	0.95	102.64	Poland	20	1.01	69.94
Belgium	20	1.13	93.14	France	20	1.13	56.50	France	21	1.05	69.87

The degree centrality, closeness centrality, and betweenness centrality in the table are all normalized metrics calculated based on the network structure and are dimensionless variables.

Table 3
Degree of centrality of renewable energy trade networks in the EU.

2021				2022				2023			
country	degree	Closeness	Betweenness	country	degree	Closeness	Betweenness	country	degree	Closeness	Betweenness
EU Renewable Energy Trade Network											
United States	15	0.72	235.76	United States	26	0.10	145.40	United States	20	0.95	157.12
China	20	0.72	181.47	China	41	1.07	155.86	China	37	0.96	137.69
Russia	0	0.33	0	Russia	1	0.49	0	Russia	0	0.33	0
Germany	12	0.73	254.07	Germany	24	1.09	207.20	Germany	26	1.02	241.10
Poland	4	0.62	39.65	Italy	24	0.97	169.78	Netherlands	22	0.97	143.97
Italy	8	0.63	38.77	France	23	1.01	145.38	France	23	0.94	129.58
Netherlands	5	0.64	31.48	Netherlands	22	0.10	130.25	Italy	22	0.92	95.50
United Kingdom	4	0.65	14.63	Lithuania	8	0.84	82.53	Lithuania	8	0.81	72.70
France	5	0.66	10.49	Austria	19	0.92	63.95	Sweden	13	0.82	40.29

The degree centrality, closeness centrality, and betweenness centrality in the table are all normalized metrics calculated based on the network structure and are dimensionless variables.

renewable energy trade network from 2021 to 2023. Despite a similar decline in Russia's out-degree, China's out-degree has increased to 41, the US has risen to 26, and the EU countries' in-degree has surged by about three times. This reflects the EU's aggressive expansion of renewable trade cooperation with the US and China in response to shortages of conventional energy sources. In particular, Germany, France, the Netherlands, Spain, and Belgium have become important import markets for Chinese PV products.

3.2.3. Core-edge structure

We divide the network into the core and the edge based on the proximity between nodes. Nodes located in the core play a crucial role and are the key economies that drive the continuous evolution of the entire network (Li et al., 2024). We introduce the core degree to quantify how active countries are in trade. The higher the core degree, the more significant the country's position in the trade. Using the Core/Periphery program of the Ucinet software, we calculated the core degree of each country, which in turn classified core (core degree ≥ 0.12) and peripheral countries (core degree < 0.12).

The core-edge structure is particularly well-characterized in the two main energy networks of the EU, and the structural relationship between the two shows a high degree of stability. Table 4 presents the core-periphery classification results for various countries from 2021 to 2023, reflecting the dynamic evolution of core and peripheral nations. The core countries remain roughly at 20, while the number of edge countries stays at 30. Notably, countries such as the US, China, the United Kingdom, France, and Germany are firmly at the core of both trade networks. The position of countries such as Poland and Sweden is more visible in the conventional energy network due to their long-standing high dependence on Russian fossil fuels (Martínez-García et al.,

2023).

However, this pattern has changed significantly since the Russia-Ukraine conflict outbreak in 2022. In the EU's conventional energy trade network, Russia has shifted from a core country to a peripheral one, reflecting its weakening influence on the EU energy supply. In the renewable energy trade network, there has been a positive shift. In addition to countries that were already in the core circle, such as Germany, countries such as Denmark, Belgium, Slovenia, Hungary, the Czech Republic, Croatia, and Israel have made the leap from the periphery to the core, marking a significant raise in their participation in the renewable energy trade sector and thereby, actively contributing to the energy transition.

4. Model construction and mechanism analysis

4.1. Model construction

The core of ERGM, which is built on various network statistics, portrays the probabilistic mechanism of network construction and the reasons behind it through an exponential function. The model is fitted using maximum likelihood estimation and undergoes multiple stages of analysis, simulation, and diagnosis to ensure accurate calibration of the model parameters and ultimately reaches a converged empirical estimation (Wang et al., 2024). The significant advantage of TERGM is the introduction of the time dimension; which enables the exploration of the deeper mechanisms of network dynamic evolution (Li et al., 2024; Wu et al., 2020).

The unique feature of ERGM and TERGM is that they can identify and incorporate interdependencies among variables as explanatory variables, effectively circumventing the limitations of traditional

Table 4

Core-periphery countries of conventional and renewable energy trade networks in the EU.

	EU Conventional Energy Trade Network		EU Renewable Energy Trade Network	
	Core countries	periphery countries	Core countries	periphery countries
2021		DNK, SAU, CZE, ARE, SGP, HRV, SVK, CHE, IDN, BGR, PRT, JPN, ROU, THA, BEL, POL, GRC, SVN, MEX, FIN, AUS, HUN, SWE, BRA, IRL, ISR, AUT, CAN, LVA, LUX, RUS, TUR, MLT, EST, LTU, ARG, CYP		BEL, CAN, SWE, THA, TUR, CZE, AUT, ISR, DNK, FIN, RUS, ARE, HUN, BGR, AUS, ROU, MLT, SAU, ARG, LVA, GRC, IRL, CYP, PRT, LUX, EST, BRA, SVK, SVN, HRV, LTU
		RUS, DNK, IDN, ROU, PRT, GRC, SVK, BGR, SAU, ARE, FIN, IND, SVN, SGP, CYP, MEX, LUX, LVA, ISR, ARG, IRL, MLT, LTU, JPN, THA, EST, BGR, FIN, AUS, BRA, CAN, GRC, CHE, RUS, LTU, TUR, ISR, SVK, CHE, GBR, IND, EST, SVN, SAU, SGP, ARE, IDN, LUX, CAN, DNK, HUN, CZE, HRV, ISR	DEU, ITA, NLD, FRA, USA, CHN, GBR, AUT, JPN, SGP, CHE, CAN, ESP, CZE, KOR, BEL, SVN, ISR, DNK, THA, HUN, HRV	LTU, ROU, PRT, BGR, GRC, SWE, IND, TUR, MEX, SVK, POL, CYP, IDN, LUX, AUS, LVA, FIN, BRA, EST, MLT, ARE, SAU, ARG, RUS, IRL
		DEU, ITA, NLD, ESP, USA, BEL, FRA, GBR, POL, CHN, SWE, AUT, KOR, HRV, CHE, HUN, CAN, CZE, AUS, BRA	DEU, NLD, ITA, FRA, USA, CHN, GBR, AUT, JPN, SGP, CHE, GBR, ESP, JPN, AUT, SGP, BEL, SVN, CAN, DNK, HUN, CZE, HRV, ISR	ROU, LTU, SWE, MEX, GRC, BGR, POL, THA, LUX, TUR, IND, KOR, IDN, PRT, AUS, CYP, SVK, FIN, LVA, IRL, ARE, EST, RUS, ARG, BRA, SAU, MLT
		ITA, DEU, NLD, GBR, FRA, BEL, ESP, POL, CZE, USA, AUT, HUN, DNK, SWE, HRV, CHN, PRT, ROU	DEU, NLD, ITA, FRA, USA, CHN, CHE, GBR, ESP, JPN, AUT, SGP, BEL, SVN, CAN, DNK, HUN, CZE, HRV, ISR	
2022		DEU, ITA, NLD, ESP, USA, BEL, FRA, GBR, POL, CHN, SWE, AUT, KOR, HRV, CHE, HUN, CAN, CZE, AUS, BRA	DEU, ITA, NLD, FRA, USA, CHN, GBR, AUT, JPN, SGP, CHE, CAN, ESP, CZE, KOR, BEL, SVN, ISR, DNK, THA, HUN, HRV	LTU, ROU, PRT, BGR, GRC, SWE, IND, TUR, MEX, SVK, POL, CYP, IDN, LUX, AUS, LVA, FIN, BRA, EST, MLT, ARE, SAU, ARG, RUS, IRL
		ITA, DEU, NLD, GBR, FRA, BEL, ESP, POL, CZE, USA, AUT, HUN, DNK, SWE, HRV, CHN, PRT, ROU	DEU, NLD, ITA, FRA, USA, CHN, CHE, GBR, ESP, JPN, AUT, SGP, BEL, SVN, CAN, DNK, HUN, CZE, HRV, ISR	
2023		ITA, DEU, NLD, GBR, FRA, BEL, ESP, POL, CZE, USA, AUT, HUN, DNK, SWE, HRV, CHN, PRT, ROU	DEU, NLD, ITA, FRA, USA, CHN, CHE, GBR, ESP, JPN, AUT, SGP, BEL, SVN, CAN, DNK, HUN, CZE, HRV, ISR	LTU, ROU, PRT, BGR, GRC, SWE, IND, TUR, MEX, SVK, POL, CYP, IDN, LUX, AUS, LVA, FIN, BRA, EST, MLT, ARE, SAU, ARG, RUS, IRL

approaches (He et al., 2019). These models comprehensively consider the multiple forces driving network evolution; analyzing how external dynamics shape the competitive landscape and examining the potential impact of structural factors within the network (Jang and Yang, 2022).

This paper uses the TERGM model given its superiority to construct two major trade networks of conventional and renewable energy in the EU from 2014 to 2023. Network endogenous structure, node attributes, and exogenous networks are incorporated into the model (Tolochko and Boomgaarden, 2024). The model is shown in eq. (1). $P_r(Y = y|\theta)$ is an explanatory variable quantifying the probability that network y is realized given the parameter θ . y refers to the conventional and renewable energy trade network of the EU. Edges represent the edges of the trade network, equivalent to the benchmark term in the regression analysis. Mutual embodies reciprocity in the network. f stands for the node attribute effect, which reveals the influence of different countries' characteristics on the network structure. r reflects the exogenous relationship between countries that drives the evolution of the network. By testing the θ , we can assess the impact of these three statistics in shaping the network morphology and evolution. In addition, $1/c$ as a standardization constant ensures the reasonableness of the probability

distribution of the model results.

$$P_r(Y = y|\theta) = (1/c) \exp\{(\theta_0 \text{edges} + \theta_1 \text{mutual}) + (\theta_{f1} \text{risk} + \theta_{f2} \ln \text{GDP} + \theta_{f3} \text{tradeopen} + \theta_{f4} \text{fdi} + \theta_{f5} \text{property} + \theta_{f6} \text{pop} + \theta_{f7} \text{urban}) + (\theta_{r1} \text{border} + \theta_{r2} \text{language} + \theta_{r3} \text{fta})\} \quad (1)$$

4.2. Measurement of variables

4.2.1. Explained variable

To define the object of study, we determined the specific classification of each type of energy product based on the Harmonized System of Customs Codes (HS codes), including coal (2701), oil (2709), natural gas (27111), wind turbines (850231), and photovoltaic cells (854,142, 854143, 854149, 854150) (Guan and An, 2017; Zhu et al., 2021; Zheng et al., 2022). We selected the 27 countries of the EU.¹ Significant energy trading countries in the world are mainly countries with high GDP, which also reflects the economic and trade strength of a country, so the top 30 economies were additionally included as the major participants in energy trade based on the GDP ranking.² Therefore, the final study covered 47 countries. Given the differences in the quality of energy products between countries but to ensure comparability between data, we use trade values in US dollars to measure trade relations between countries, thus eliminating interference due to fluctuations in currency exchange rates (Zhang et al., 2024a). We use import data to represent the actual trade value between countries since import data are usually reported more accurately due to stricter customs regulations on the import chain (Zheng et al., 2022).

This paper's core explanatory variable is the probability of constructing the EU trade network between conventional and renewable energy sources. We identify the existence of a trade relationship between two countries when the country's imports of conventional or renewable energy to country j reach or exceed the trade volume threshold, in which case $P_r(Y = y|\theta) = 1$, otherwise 0. The data are obtained from the UN Comtrade.

4.2.2. Explanatory variables

The model's explanatory variables cover endogenous network structure, node attributes, and exogenous networks. Given that the introduction of triangular structural variables in a highly aggregated network environment may lead to the problem of model degradation and parameter dispersion (Hunter et al., 2008), it was found to be ineffective after several model debugging exercises. Therefore, the edges and mutual were selected as endogenous structural variables. The edges measure the probability of a trade relationship occurring when all else is equal. Mutual reveals the degree of active trade between countries in the network, and its significant positive value predicts the existence of close two-way trade between nations.

The nodal attribute effect reflects the tendency to form trade between countries with a particular attribute and other countries. By introducing nodecov parameters, we aim to explore how economic and political factors affect these trades (Jang and Yang, 2024). We incorporate variables such as geopolitical risk index (risk); gross domestic product (GDP); trade openness (trade open); foreign investment level (FDI); intellectual property level (property); population size (pop); and degree of urbanization (urban). The GPR index constructed by Caldara and Iacoviello meticulously classifies eight categories of events by counting the number of news articles reporting on tensions. It thus

¹ The United Kingdom left the European Union in 2021, but it ranked among the top 30 economies globally by GDP. Therefore, the UK is included throughout the entire study period.

² <https://www.bp.com/content/dam/bp/business-sites/en/global/corporate/pdfs/energy-economics/statistical-review/bp-stats-review-2022-full-report.pdf>

monthly quantifies the level of geopolitical risk in 44 countries (Caldara and Iacoviello, 2022; Abdullah et al., 2024).

On the other hand, the country risk index from the International Country Risk Guide (ICRG) has been widely adopted as another essential tool for measuring a country's political risk (Tang et al., 2021; Berggren et al., 2009). The index covers annual risk data for 154 countries worldwide between 1984 and 2022, especially its political risk component, which combines 12 key indicators such as external conflict, internal conflict, and ethnic and religious tensions (Hammoudeh et al., 2013). The ICRG index is relatively superior regarding the range of countries covered and the diversity of indicators assessed (Hoti and Mcaleer, 2004; Lee and Hooy, 2012). Given this, the risk indicators in ICRG were selected for this study to comprehensively reflect countries' political risk (Chen et al., 2016). It is worth noting that the index uses reverse scoring; i.e., a low score corresponds to high risk; and a high score corresponds to low risk (Aziz, 2018). The raw risk values have been transformed in this paper so that an increase in risk values directly corresponds to a rise in risk levels. All other data indicators in this study are derived from the World Bank.

Network covariates analyze the contribution of external network relationships to the formation of the overall network. We focus on the contribution of external network relationships such as geography, language, and free trade agreements to forming the overall network. Among them, geographic proximity (border) and linguistic commonality (language), as static network variables, reflect long-term stable ties between countries. A free trade agreement (FTA) is a dynamic network variable that reflects dynamic economic and trade relations between nations. We construct a binary matrix based on the CEPII and WTO database, where 1 indicates a corresponding external link between the two countries and 0 otherwise. Table 4 details the descriptions of the variables and the data sources.

4.3. Sample and data sources

The origins of the Russo-Ukrainian conflict can be traced back to the 2014 crisis in Ukraine and Crimea, which led to domestic political unrest in Ukraine and triggered Russia's military intervention. Therefore, this study selects 47 major energy-trading countries between 2014 and 2023 as the research object to comprehensively examine the evolutionary characteristics of the EU's energy trade network and its drivers during this period. We take the following steps in data organization and pre-processing:

First, referring to the study of (Desmarais and Cranmer, 2012), we set a threshold value of energy trade volume at one million dollars to

highlight key trade links and reduce extreme values in the data. We converted the original energy trade matrix into a binary matrix and included it as an explanatory variable in ERGM.

Second, we log-transform some continuous variables to reduce the possible impact of the heteroskedasticity problem on the analysis, including the node attribute variables such as Gross National Product (GDP), Foreign Direct Investment (FDI) level, Intellectual Property (IP) level, and Population size (Pop).

Finally, we manually screen and code the corresponding information on Free Trade Agreements (FTA) signing between countries from the WTO database. Specifically, if an FTA has been signed between two countries, a value of 1 is assigned to the corresponding adjacency matrix, otherwise 0, to ensure that the discrete treatment of the variable meets the requirements set by the model. Table 5 lists the primary variables used in the ERGM, including their names, definitions, and data sources.

4.4. Results

4.4.1. Static analysis

We use stepwise regression to construct a comprehensive estimation model to explore the effects of the variables on the EU trade network between conventional and renewable energy sources. This modeling process starts with a baseline model containing only endogenous structure, followed by the gradual inclusion of node attribute effects to test the role of exogenous mechanisms. Exogenous network covariates are then added, and geopolitical risk factors are eventually integrated to construct a test model that incorporates the effects of both internal and external mechanisms. Table 6 presents the integrated model results for the EU's two major energy trade networks from 2021 to 2023. The results show that the AIC and BIC values continue to decrease as the model complexity increases, indicating that the model that considers the endogenous structure, node attributes, and exogenous network is superior.

In terms of endogenous structure for the EU traditional energy trade network; the negative value of the edges parameter indicates non-random formation of trade relations, validating the importance of studying the mechanisms of network formation and evolution. The mutual coefficient was significantly positive before the Russia-Ukraine conflict. Then it weakened, reflecting that the conflict weakened reciprocal trade, the EU's reduced dependence on Russian energy, and that the sanctions further restricted trade flows and accelerated the restructuring of supply chains. The effect of geopolitical risk is significantly negative in the post-conflict period, suggesting that political

Table 5
Description of the variables of ERGM.

	Variable Name	Definition and Measurement	Data sources
Dependent Variable	$P_r(Y = y \theta)$	Binary variable for trade volume between the EU and 47 other countries in conventional and renewable energy: 1 if exceeding \$1 million, 0 otherwise	UN Comtrade
Network Structure	Edges	Network edge counts, functioning similarly to constant terms, are generally not interpreted.	Ucinet
	Mutual	The closeness of bilateral trade relations	
	Risk	Geopolitical Risk Twelve key indicators encompassing external conflicts, internal conflicts, ethnic and religious tensions, etc.	ICRG
	GDP	Gross Domestic Product Measures the total value added created by a country's residents and producers, converted at official exchange rates and expressed in current US dollars.	
	Trade open	Trade openness The ratio of total goods exports and imports to GDP, expressed in current US dollars.	
Node Attributes	FDI	Foreign Direct Investment Net capital inflows resulting from foreign investors holding over 10% of voting rights, measured in current US dollars.	World Bank
	Property	Intellectual Property Receipts and payments for royalties and license fees arising from the use of intangible assets such as patents and copyrights, measured in current US dollars.	
	Urban	Urbanization Level The proportion of the urban population relative to the total population.	
	Pop	Population The total resident population of a country, estimated at mid-year.	
	Border	Common border 1 if countries are neighbors, 0 otherwise.	
Network Covariates	Language	Common language 1 if languages are identical, 0 otherwise.	CEPII
	FTA	Free Trade Agreement 1 if signed, 0 otherwise.	WTO

Table 6
Results of static analysis based on ERGM, 2021–2023.

	Conventional Energy Trade Network			Renewable Energy Trade Network		
	2021	2022	2023	2021	2022	2023
edges	−21.482*** (2.490)	−22.196*** (2.561)	−18.017*** (2.827)	−62.001*** (7.024)	−29.797*** (3.170)	−25.031*** (3.031)
mutual	0.742*** (0.180)	0.688*** (0.173)	0.052 (0.192)	1.270*** (0.373)	0.673*** (0.193)	0.887*** (0.189)
risk	−0.001 (0.006)	−0.007 (0.005)	−0.017** (0.006)	0.012 (0.009)	0.014* (0.006)	0.000 (0.006)
GDP	0.131 (0.092)	0.198* (0.094)	0.065 (0.107)	0.928*** (0.256)	0.432*** (0.126)	0.289* (0.117)
trade open	0.003** (0.001)	0.004*** (0.001)	0.002* (0.001)	0.010*** (0.002)	0.007*** (0.001)	0.008*** (0.001)
FDI	−0.010 (0.008)	0.008 (0.005)	−0.005 (0.008)	0.013 (0.014)	0.036*** (0.006)	−0.006 (0.007)
property	0.017 (0.010)	0.030** (0.009)	0.038*** (0.010)	0.168* (0.076)	0.254*** (0.042)	0.180*** (0.032)
pop	0.290*** (0.086)	0.200* (0.088)	0.282** (0.099)	−0.055 (0.190)	−0.208* (0.096)	−0.002 (0.096)
urban	0.014*** (0.004)	0.009* (0.008)	0.016*** (0.004)	−0.004 (0.007)	−0.016*** (0.004)	−0.006 (0.004)
border	2.950*** (0.295)	2.518*** (0.292)	2.897*** (0.286)	1.240*** (0.352)	1.593*** (0.261)	1.704*** (0.248)
language	−0.161 (0.236)	−0.125 (0.22)	−0.316 (0.248)	−0.025 (0.351)	0.161 (0.238)	0.261 (0.228)
fta	0.500* (0.253)	0.012 (0.834)	1.066 (1.027)	0.270 (0.424)	0.368 (0.862)	0.000 (0.000)
AIC	1909.896	2036.529	1977.528	607.984	1750.509	1721.225
BIC	1978.041	2104.675	2045.674	676.129	1818.655	1783.692

***, **, and * indicate statistical significance at 1%, 5%, and 10% levels, respectively; the values in parentheses represent z-statistics.

instability increases trade uncertainty and that firms and governments reduce their reliance on conflict regions. European countries, at the same time, increased imports from other suppliers and sought long-term energy independence, affecting the stability of traditional energy trade. GDP, trade openness, intellectual property rights level, population, and urbanization level consistently positively impact trade, suggesting that it is easier to establish stable energy trade relations in countries with stronger economies and more open markets, among others. Thus, **H1** is verified. Geographic proximity and free trade agreements significantly promote trade. Neighboring countries have smoother energy flows due to low transportation costs and infrastructure interoperability. At the same time, free trade agreements reduce barriers, promote long-term stable cooperation, and facilitate the expansion of trade networks. Hence, **H2** is verified.

In terms of endogenous structure for the EU renewable energy trade network, edges are equally harmful, emphasizing the existence of non-random trade relations. The coefficient of mutual is significantly positive, indicating that reciprocal trade relations are more stable and less affected by geopolitical conflicts. The effect of geopolitical risk is consistently positive and significantly promotes renewable energy trade in the aftermath of conflict. The Russia-Ukraine conflict has reduced the EU's dependence on Russia for traditional energy sources. At the same time, the limited global supply has made it difficult to bridge the gap, prompting countries to accelerate the development of photovoltaic and solar energy to improve energy security. This trend has boosted regional renewable energy investments and accelerated technology diffusion and market consolidation. GDP, trade openness, and intellectual property rights levels contribute significantly to renewable energy trade, suggesting that economic and technological advances are critical to driving the renewable energy market. The negative impact of population and urbanization levels may stem from the fact that highly urbanized countries are more energy-autonomous and can rely on their own solar and photovoltaic products to meet demand and, therefore, are less dependent on imports. In the aftermath of conflict, there is a greater focus on enhancing their production capacity to reduce the impact of external market volatility. Therefore, **H1** is verified. Both geographic proximity and free trade agreements significantly promote trade with

neighboring countries are more likely to form cross-border energy cooperation while policy coordination further promotes green energy market development. Hence, **H2** is verified.

As shown in Fig. 4, we implement a simulation test based on the Goodness of Fit (GOF) to assess the applicability of the ERGM model in portraying the EU energy trade network. Specifically, Fig. 4(a) depicts the results of the GOF test for the conventional energy trade network in 2023, while Fig. 4(b) corresponds to the test for the renewable energy trade network in 2023. The model-fitting effect can be judged by comparing the similarity in structural features between the simulation-generated analog network and the actual network (Jang and Yang, 2024).

This paper selected and analyzed six key metrics, with 10,000 fits for each. The first group includes the degree of sharing, the number of binary sharing partners, and the geodesic distance. In contrast, the second group covers the degree, the ternary group census, and the receiver operating characteristic curve (ROC). The ROC curve assesses the model's ability to recognize trade relationships by comparing the accuracy between real trade presence and simulated predictions. The vertical axis is the frequency, the solid line is the structural characteristics of the actual model, and the box plots represent the structural characteristics of the model obtained from the simulation.

As shown in Fig. 4, the integrated model fits well in terms of the proximity of the box plots' middle values to the actual model's curve in the first five subplots and the bias of the ROC curve to the upper left. This finding is consistent with the results obtained when assessing the model's quality through the AIC and BIC values, reinforcing the robustness of the previous analysis. These findings not only emphasize the critical role of geopolitical risks in energy trade but also point to the limitations of relying solely on endogenous structural mechanisms for analysis and stress the importance of integrating exogenous mechanisms to gain a more comprehensive and in-depth understanding of the changes of the EU's energy trade network and the complex dynamics behind it.

4.4.2. Dynamic analysis

Given the limitations of static data analysis, it is difficult for the

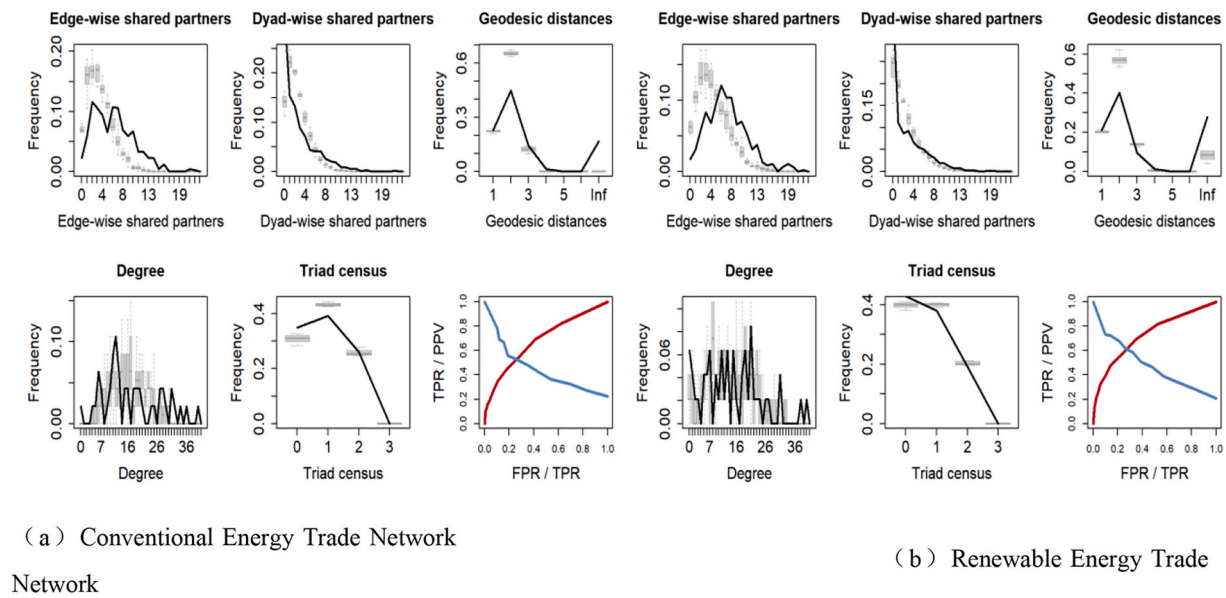


Fig. 4. Goodness-of-fit test in the static model for conventional and renewable energy trade networks.

ERGM model to adequately reveal the dynamic evolution of the EU's conventional and renewable energy trade networks. For this reason, we integrated the data from 2014 to 2023 and used the TERGM model for stepwise regression to re-estimate the parameters in Eq. (1). The model construction process started from the baseline model 1, which only considered endogenous structural effects, and was gradually extended to model 2, which incorporated the effects of node attributes to explore the impacts of exogenous mechanisms. Subsequently, Model 3 integrates exogenous network covariates to realize the initial integration of internal and external mechanisms. Finally, Model 4 introduces geopolitical risk factors to form a comprehensive, integrated test model. Table 7 presents the estimation results for each model from 2014 to 2023 which

is used to analyze the impact of endogenous structure, node attributes, and exogenous network covariates on the dynamic evolution of trade networks. The results show that the AIC and BIC values continue to decrease as the model complexity increases, verifying the superiority in the explanatory power of the model that simultaneously considers the endogenous structure, the node attributes, and the exogenous networks.

The negative values of the edges parameter in each model consistently indicate that constructing the EU's energy trade network is not a random process but instead is driven by a combination of economic, political, and policy factors across countries, which underscores the importance of examining the mechanisms of its formation and evolution. The significant positive value of the mutual coefficient reveals the

Table 7

Results of dynamic analysis based on TERGM, 2014–2023.

	Conventional Energy Trade Network			Renewable Energy Trade Network		
	model1	model2	model3	model1	model2	model3
edges	-1.630*** (0.022)	-23.294*** (0.680)	-23.344*** (0.754)	-2.939*** (0.035)	-39.650*** (1.335)	-41.086*** (1.371)
mutual	1.482*** (0.050)	0.594*** (0.057)	0.595*** (0.057)	2.941*** (0.078)	1.624*** (0.091)	1.613*** (0.090)
risk			0.000 (0.001)			0.005** (0.002)
lnGDP		0.224*** (0.022)	0.227*** (0.026)		0.597*** (0.047)	0.653*** (0.049)
trade open		0.003*** (0.000)	0.003*** (0.000)		0.010*** (0.000)	0.010*** (0.000)
FDI		-0.004* (0.002)	-0.004* (0.002)		0.016*** (0.002)	0.016*** (0.003)
property		0.014*** (0.003)	0.014*** (0.003)		0.141*** (0.012)	0.145*** (0.013)
pop		0.203*** (0.019)	0.199*** (0.023)		-0.074* (0.033)	-0.129*** (0.037)
urban		0.012*** (0.001)	0.012*** (0.001)		-0.009*** (0.002)	-0.010*** (0.002)
border		2.804*** (0.092)	2.806*** (0.091)		0.936*** (0.087)	0.922*** (0.088)
language		-0.158* (0.076)	-0.160* (0.073)		-0.081 (0.087)	-0.054 (0.086)
fatalist		1.260*** (0.320)	1.294*** (0.316)		1.490*** (0.330)	1.403*** (0.330)
AIC	29,359.362	19,006.472	19,007.452	13,233.463	9025.277	9019.883
BIC	29,379.968	19,119.809	19,131.092	13,254.069	9138.614	9143.523

***, **, and * indicate statistical significance at 1%, 5%, and 10% levels, respectively; the values in parentheses represent z-statistics.

strengthening of the bilateral trade relationship, suggesting that inter-country energy cooperation is gradually deepening and evolving into a more stable and lasting partnership. This trend may stem from the rising demand for energy security, prompting countries to reduce energy supply risks through long-term cooperation while optimizing the cross-border allocation of energy resources.

Geopolitical risks exert a greater and more significant impact on renewable energy trade networks, indicating that geopolitical shocks play a stronger role in driving the expansion of EU renewable energy trade than they do in conventional energy trade. GDP and trade openness exhibit a significant positive relationship in both types of energy trade networks, suggesting that economic scale and trade openness are crucial factors influencing a country's energy imports. However, the effects of foreign investment, intellectual property rights, and population and urbanization levels on the two types of energy trade are divergent. The promotion of renewable energy trade by the level of foreign investment and intellectual property rights suggests that renewable energy technologies are dependent on capital and technological inputs and that a high level of foreign investment and intellectual property rights protection can accelerate the research and development and diffusion of green energy technologies, thereby promoting international trade. The positive impact of population and urbanization levels on trade in traditional energy sources may reflect that countries with high population densities and a high degree of urbanization still maintain a strong dependence on conventional energy sources, especially in terms of transportation, manufacturing, and residential energy demand. This dependence may gradually weaken in the future as renewable energy technologies advance and policy incentives increase. These findings, therefore, support *H1*.

The positive coefficient of geographic proximity and FTAs reveals the critical role of geographic proximity and trade cooperation in facilitating the establishment of trade relations. Geographic proximity reduces transportation costs and improves the ease of energy flow, making energy trade between neighboring countries more attractive. In addition, the significant positive effect of free trade agreements suggests that policy coordination and market openness can further eliminate trade barriers and enhance the stability of energy supply chains. This finding reinforces the fundamental role of geography in international trade and validates the potential impact of institutional cooperation in energy trade, supporting *H2*.

We executed 10,000 simulations using the model 3 parameters in Table 7 for the two networks. As shown in Fig. 5, Fig. 5(a) focuses on the

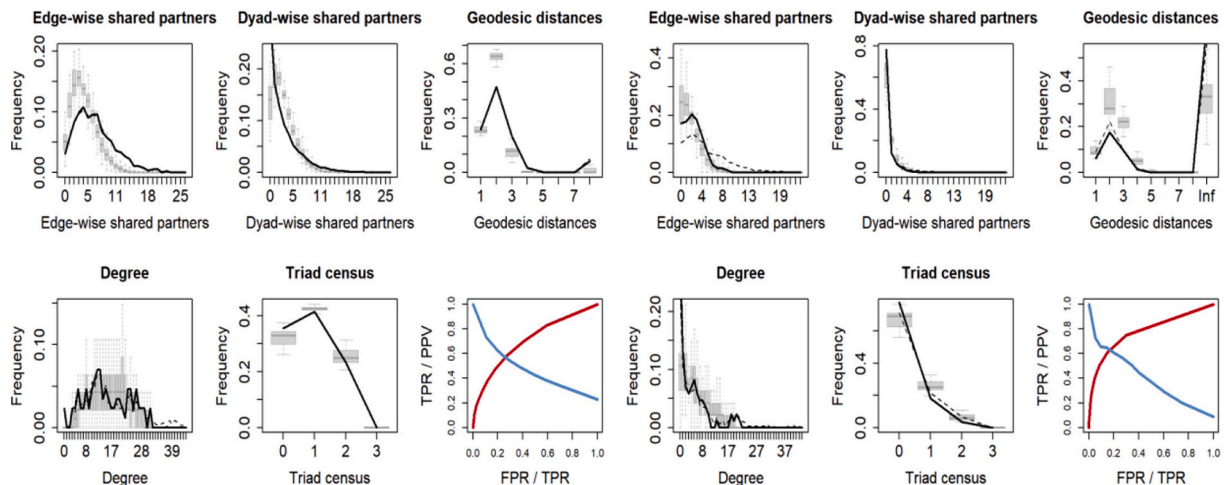
goodness-of-fit test for the conventional energy trade network under the dynamic model. At the same time, Fig. 5(b) corresponds to the test for renewable energy. In both figures, the accuracy of the integrated model fit is reflected in the closeness of the middle of the box to the actual model curve in the first five subfigures and the degree to which the ROC curve is biased to the upper left. This finding is consistent with the conclusions reached through the AIC and BIC analyses, further solidifying the robustness of the previous tests. It highlights the significant role of geopolitical risk factors in energy trade while emphasizing the limitations of a study that relies solely on endogenous structural mechanisms.

4.4.3. Robustness test

Table 8 presents the results of robustness tests conducted to verify whether TERGM estimation conclusions remain consistent under different settings. First, the trade threshold was adjusted from \$1 million to the annual average trade value for each year, and the model was re-estimated. The results are shown in columns (1) and (3) of Table 8. Second, considering the impact of the COVID-19 pandemic on energy trade in 2019, the sample period was shortened to 2019–2023, and the TERGM analysis was re-conducted. The results are presented in columns (2) and (4) of Table 8.

The comprehensive analysis results indicate that geopolitical risks significantly negatively impact conventional energy trade, implying that an increase in geopolitical risk levels leads to a decline in the volume of conventional energy transactions between trading partners. However, geopolitical risks exert a notable positive influence in the context of renewable energy trade, suggesting that an elevation in geopolitical risks paradoxically fosters the development of renewable energy trade. This phenomenon may be attributed to countries' proactive pursuit of energy transition pathways in the face of conflicts and uncertainties aimed at reducing dependence on conventional energy sources and enhancing energy security, such as increasing imports of renewable energy products, such as wind and solar power, to develop domestic wind and solar industries. This finding reveals the unexpected role of geopolitical risks in driving energy structure transformation and renewable energy development.

Columns (2) and (4) of Table 8 present the results of the post-epidemic analysis. Compared to the model covering all years in Table 7, the analysis using only 2019–2023 data shows that the effects of the node attributes are still significant and that geopolitical risk further strengthens the dampening impact on traditional energy trade while



(a) Conventional energy Trade Network

(b) Renewable Energy Trade Network

Fig. 5. Goodness-of-fit test in the dynamic model for conventional and renewable energy trade networks.

Table 8
Robustness test results.

	Conventional Energy Trade Network		Renewable Energy Trade Network	
	(1)	(2)	(3)	(4)
edges	-33.865 *** (2.157)	-20.224 *** (1.132)	-47.172 *** (2.473)	-39.206 *** (1.650)
mutual	-1.551 *** (0.217)	0.611 *** (0.083)	0.996 *** (0.148)	1.482 *** (0.110)
risk	-0.025 *** (0.005)	-0.009 *** (0.002)	0.010 ** (0.004)	0.015 *** (0.003)
GDP	-0.013 (0.076)	0.111 ** (0.040)	0.626 *** (0.090)	0.712 *** (0.059)
trade open	-0.001 (0.001)	0.003 *** (0.000)	0.008 *** (0.001)	0.009 *** (0.001)
FDI	-0.015 ** (0.005)	-0.005 * (0.002)	-0.001 (0.004)	0.019 *** (0.003)
property	-0.044 *** (0.007)	0.020 *** (0.004)	0.160 *** (0.026)	0.139 *** (0.015)
pop	0.833 *** (0.077)	0.288 *** (0.038)	0.050 (0.069)	-0.271 *** (0.048)
urban	0.041 *** (0.003)	0.014 *** (0.002)	-0.003 (0.003)	-0.011 *** (0.002)
border	2.064 *** (0.182)	2.795 *** (0.130)	1.296 *** (0.147)	1.048 *** (0.117)
language	0.505 ** (0.167)	-0.170 (0.108)	-0.184 (0.149)	-0.002 (0.116)
fta	0.352 (0.691)	1.482 (0.479)	0.952 (0.614)	0.985 * (0.476)
AIC	3956.345	9643.091	3768.693	5766.306
BIC	4063.324	9750.070	3875.671	5873.285

***, **, and * indicate statistical significance at 1%, 5%, and 10% levels, respectively; the values in parentheses represent z-statistics.

boosting renewable energy trade more significantly. This suggests that geopolitical risk weakens traditional energy flows and boosts renewables, reflecting an acceleration of this trend caused by the epidemic. During the epidemic, supply chain disruptions, increased volatility in fossil energy markets, and rising awareness of energy security among countries made renewable energy trade an essential means of coping with energy uncertainty. At the same time, the epidemic pushed countries to increase green investment and policy support, further accelerating the expansion of the renewable energy market and exacerbating the restructuring of the energy trade landscape.

In summary, the conclusions from the robustness analysis remain consistent with the main analysis above despite changes in the analysis thresholds and sample period which asserts the robustness of the main findings.

5. Conclusions and policy recommendations

5.1. Conclusion

The global energy market has undergone profound changes since the Russia-Ukraine conflict, and the European Union's energy security has faced unprecedented challenges. In addition to the restructuring of the supply chain due to the conflict, the global decarbonization process, energy independence strategies, and technological innovations have also contributed to the restructuring of the energy landscape. The EU's short-term efforts to reduce its dependence on a single supplier country by increasing US LNG imports, restarting some fossil energy facilities, and accelerating renewable energy investments have contributed to the decentralization of traditional energy trade and the accelerated expansion of renewable energy trade.

The study results show a clear trend of decentralization of the traditional energy trade network in the EU, with a decrease in trade links and a decline in network density. In contrast, the renewable energy trade has become more closely linked, and its density has risen significantly. This change stems from the EU's policy of diversification of energy supply, especially post-conflict reduction of dependence on Russia

and accelerated renewable energy transition. Its impact is not limited to trade restructuring, but it also reshapes the global energy market landscape and pushes international energy policy towards decentralization and greening.

Regarding network structure, the core-edge pattern has been adjusted, with Russia gradually withdrawing from the core position and the United States becoming the leading supplier to the European Union of its energy exports and geopolitical advantages. Meanwhile, Germany, France, the Netherlands, the United Kingdom, and Italy remain at the core, playing the role of trade hubs. The changing landscape is driven by supply chain reconfiguration and geopolitical games, which affect energy trade and reshape the international political landscape, such as the intensification of energy cooperation between the EU and the US and Russia's shift to Asian markets.

A combination of factors drives the evolution of energy trade networks. Reciprocity reduces transaction costs and increases supply chain resilience, while geopolitical risks undermine traditional energy trade and accelerate renewable energy development. Economic scale and trade openness facilitate energy cooperation. In addition, foreign investment, technological innovation, geographic proximity, and free trade agreements lower trade barriers and facilitate the expansion of renewable energy. Together, these factors have influenced the reshaping of energy trade networks and will continue to affect the stability and development of energy markets.

5.2. Policy recommendations

Firstly, countries should actively promote the strategy of diversifying energy supply sources to enhance energy security. The EU should actively broaden its international cooperation horizons, not only limited to traditional partners, but also deepen its energy cooperation with key regions such as the United States, the Middle East, and North Africa, to build a geographically transversal, long-term and stable, diversified energy supply chain system. At the same time, China should implement a comprehensive and in-depth energy import strategy and deepen cooperation with energy-rich regions through diversified import channels to effectively diversify the risk of ensuring the sustainability of the energy supply. On the other hand, Russia needs to be flexible and changeable, adjust its energy export strategy, reduce excessive dependence on the European market, and actively explore Asia, Africa, and other emerging markets to realize the diversified layout of export markets.

Secondly, countries need to accelerate the research and development of green energy technology and promote the green transformation of energy structure. The EU should introduce more incentives to encourage investment and construction of renewable energy, such as wind and solar, gradually reduce dependence on fossil fuels, and promote green changes in energy consumption patterns. On the other hand, China should focus on innovation and industrial upgrading of energy technology, significantly increasing R&D investment in cutting-edge technologies such as hydrogen energy and energy storage to improve the efficiency of domestic energy utilization and show strong competitiveness in the global energy arena. China, Russia, and China-Europe should strengthen cooperation in clean energy, share innovation results, and jointly lead the global energy green transformation.

Finally, in the face of increasingly complex energy security challenges, the importance of international cooperation has become increasingly prominent. Countries should uphold the concept of building a community of human destiny and regard energy security as a global shared responsibility and mission. The EU needs to expand its global energy cooperation network further, especially by utilizing the unique position of the United States as an emerging energy hub and jointly building a more solid, flexible, and efficient global energy supply chain system. China needs to strengthen its dialogue and cooperation mechanism with energy-producing and consuming countries and jointly address the volatility and geopolitical risks of the energy market through

multilateral cooperation platforms to maintain the stability and prosperity of the energy trade. On the other hand, Russia needs to enhance its awareness of geopolitical risks, actively seek in-depth cooperation with its primary energy partners in energy technology and safety, and jointly promote the improvement and transformation of the global energy governance system. By strengthening international cooperation and coordination, all countries can respond more effectively to energy security challenges, promote the sustainable development of global energy, and lay a solid foundation for global economic prosperity and social stability.

5.3. Research limitations and future directions

This study analyzes the dynamic evolution of the EU's conventional and renewable energy trade networks from a network perspective. However, certain limitations remain. First, the trade network construction employs a binary approach (using a \$1 million threshold). This method may overlook the impact of small-scale trade on network structure even though annual average thresholds were applied in robustness tests. Future research could employ continuous trade values for comparative testing using econometric methods to further assess robustness. Second, the study period spans 2014–2023, covering the Russia-Ukraine conflict and key phases of the EU's energy transition. However, its portrayal of long-term evolutionary characteristics remains limited. Extending the research timeline and integrating long-term policy and technological data could reveal sustained trends in energy trade networks.

CRediT authorship contribution statement

Shuanglei Xu: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Youyi Deng:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Rabindra Nepal:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Tooraj Jamasb:** Writing – review & editing, Validation, Supervision, Conceptualization.

Declaration of competing interest

No potential conflict of interest was reported by the authors.

Acknowledgements

The article is sponsored by “the Postgraduate Innovative Research Fund” of University of International Business and Economics (Grant No.202581). The authors have contributed equally to this paper. Tooraj Jamasb acknowledges support from CSEI funded by CBS and industrial partners.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eneco.2026.109332>.

Data availability statement

The data supporting this study's findings are available from the corresponding author upon reasonable request.

References

Abdullah, M., Tiwari, A.K., Hossain, M.R., et al., 2024. Geopolitical risk and firm-level environmental, social, and governance (ESG) performance. *J. Environ. Manag.* 363, 121245.

- Acheampong, A.O., Boateng, E., Amponsah, M., et al., 2021. Revisiting the economic growth–energy consumption nexus: does globalization matter? *Energy Econ.* 102, 105472.
- Alam, M.M., Murad, M.W., 2020. The impacts of economic growth, trade openness, and technological progress on renewable energy use in organizations for economic cooperation and development countries. *Renew. Energy* 145, 382–390.
- Antimiani, A., Costantini, V., Pagliarunga, E., 2023. Fossil fuels subsidy removal and the EU carbon neutrality policy. *Energy Econ.* 119, 106524.
- Apeti, A.E., N'doua, B.D., 2023. The impact of timber regulations on timber and timber product trade. *Ecol. Econ.* 213, 107943.
- Arndt, C., 2023. Climate change vs energy security? The conditional support for energy sources among western Europeans. *Energy Policy* 174, 113471.
- Asghar, M., Ali, S., Hanif, M., et al., 2024. Energy transition in newly industrialized countries: a policy paradigm in the perspective of technological innovation and urbanization. *Sustain. Fut.* 7, 100163.
- Astrov, V., Ghodsi, M., Grieveson, R., et al., 2022. Russia's invasion of Ukraine: assessment of the humanitarian, economic, and financial impact in the short and medium term. *IIEP* 19, 331–381.
- Aziz, O.G., 2018. Institutional quality and FDI inflows in Arab economies. *Financ. Res. Lett.* 25, 111–123.
- Batzella, F., 2024. Slowly but surely? Assessing EU actorness in energy sanctions against Russia. *Energy Policy* 192, 114233.
- Bazilian, M., Cuming, V., Kenyon, T., 2020. Local-content rules for renewables projects don't always work. *Energy Strat. Rev.* 32, 100569.
- Berggren, N., Bergh, A., Björnskov, C., 2009. The Growth Effects of Institutional Instability. Lund University, Department of Economics (Working Papers 8.).
- Blondeil, M., Bradshaw, M.J., Bridge, G., et al., 2021. The geopolitics of energy system transformation: a review. *Geogr. Compass* 15, e12580.
- Cai, X., Liu, C., Zheng, S., et al., 2023. Analysis on the evolution characteristics of barite international trade pattern based on complex networks. *Res. Policy* 83, 103593.
- Cai, Y., Wu, Y., 2021. Time-varying interactions between geopolitical risks and renewable energy consumption. *Int. Rev. Econ. Financ.* 74, 116–137.
- Cai, Y., Zhang, D., Chang, T., et al., 2022. Macroeconomic outcomes of OPEC and non-OPEC oil supply shocks in the euro area. *Energy Econ.* 109, 105975.
- Caldara, D., Iacoviello, M., 2022. Measuring geopolitical risk. *Am. Econ. Rev.* 112, 1194–1225.
- Carodenuto, S., Ziga-Abortta, F.R., Sotirov, M., 2024. External Europeanization through timber trade agreements: Tracing causality in environmental governance reform. *Polit. Geogr.* 109, 103065.
- Chen, H., Liao, H., Tang, B.-J., et al., 2016. Impacts of OPEC'S political risk on the international crude oil prices: An empirical analysis based on the SVAR models. *Energy Econ.* 57, 42–49.
- Chen, Y., Jiang, J., Wang, L., et al., 2023. Impact assessment of energy sanctions in geo-conflict: Russian–Ukrainian war. *Energy Rep.* 9, 3082–3095.
- Desmarais, B.A., Cranmer, S.J., 2012. Statistical mechanics of networks: estimation and uncertainty. *Phys. A-Statist. Mech. Appl.* 391, 1865–1876.
- Desogus, E., Grosso, D., Bompard, E., et al., 2023. Modelling the geopolitical impact on risk assessment of energy supply system: the case of Italian crude oil supply. *Energy* 284, 128578.
- Dong, Y., Li, C., 2018. Economic sanction games among the US, the EU and Russia: payoffs and potential effects. *Econ. Model.* 73, 117–128.
- Dussaux, D., Dechezleprêtre, A., Glachant, M., 2022. The impact of intellectual property rights protection on low-carbon trade and foreign direct investments. *Energy Policy* 171, 113269.
- Estrada, M.A.R., Koutronas, E., 2022. The impact of the Russian aggression against Ukraine on the Russia-EU trade. *J. Policy Model* 44, 599–616.
- Feng, L., Xu, H., Wu, G., et al., 2021. Service trade network structure and its determinants in the belt and road based on the temporal exponential random graph model. *Pac. Econ. Rev.* 26, 617–650.
- Filipović, S., Lior, N., Radovanović, M., 2022. The green deal – just transition and sustainable development goals nexus. *Renew. Sust. Energy Rev.* 168, 112759.
- Freeman, L.C., 1979. Centrality in social networks conceptual clarification. *Soc. Networks* 1, 215–239.
- Gao, B., Xu, Z., 2024. Adjustment of supply chain investment by transnational corporations in the Ukrainian crisis and geopolitical risks – taking the three levels as the analytical framework. *Transnatl. Corp. Rev.* 16, 200076.
- Gao, C., Sun, M., Shen, B., 2015. Features and evolution of international fossil energy trade relationships: a weighted multilayer network analysis. *Appl. Energy* 156, 542–554.
- Gong, X., Sun, Y., Du, Z., 2022. Geopolitical risk and China's oil security. *Energy Policy* 163, 112856.
- Gong, X.-L., Feng, Y.-K., Liu, J.-M., et al., 2023. Study on international energy market and geopolitical risk contagion based on complex network. *Res. Policy* 82, 103495.
- Guan, Q., An, H., 2017. The exploration on the trade preferences of cooperation partners in four energy commodities' international trade: crude oil, coal, natural gas and photovoltaic. *Appl. Energy* 203, 154–163.
- Guo, Y., Zheng, H., Zeng, Y., et al., 2023a. Natural resources extraction of RCEP trade bloc: examining geopolitical risk and economic situation. *Res. Policy* 86, 104227.
- Guo, Y., Zheng, R., Zhang, H., 2023b. Tantalum trade structural dependencies are what we need: a perspective on the industrial chain. *Res. Policy* 82, 103469.
- Hammoudeh, S., Sari, R., Uzunkaya, M., et al., 2013. The dynamics of BRICS'S country risk ratings and domestic stock markets, US stock market and oil price. *Math. Comput. Simul.* 94, 277–294.
- He, X.-j., Dong, Y.-b., Wu, Y.-y., et al., 2019. Factors affecting evolution of the interprovincial technology patent trade networks in China based on exponential random graph models. *Physica A Stat. Mech. Appl.* 514, 443–457.

- Hoti, S., McAleer, M., 2004. An empirical assessment of country risk ratings and associated models. *J. Econ. Surv.* 18, 539–588.
- Hunter, D.R., Goodreau, S.M., Handcock, M.S., 2008. Goodness of fit of social network models. *J. Am. Stat. Assoc.* 103, 248–258.
- Jang, Y., Yang, J.-S., 2022. The dynamics of the EU'S nuclear trade network: An ERGM analysis. *Struct. Chang. Econ. Dyn.* 63, 470–477.
- Jang, Y., Yang, J.-S., 2024. Environmental policy and the evolution of nuclear trade network: insights from the European Union. *Struct. Chang. Econ. Dyn.* 68, 425–432.
- Jawadi, F., Rozin, P., Gnegne, Y., et al., 2024. Geopolitical risks and business fluctuations in Europe: a sectorial analysis. *Eur. J. Polit. Econ.* 85, 102585.
- Jiao, J.-W., Yin, J.-P., Xu, P.-F., et al., 2023. Transmission mechanisms of geopolitical risks to the crude oil market - a pioneering two-stage geopolitical risk analysis approach. *Energy* 283, 128449.
- Khan, K., Khurshid, A., Cifuentes-Faura, J., 2023a. Energy security analysis in a geopolitically volatile world: a causal study. *Res. Policy* 83, 103673.
- Khan, K., Khurshid, A., Cifuentes-Faura, J., 2023b. Is geopolitics a new risk to environmental policy in the European Union? *J. Environ. Manag.* 345, 118868.
- Khan, M.T.I., Yaseen, M.R., Ali, Q., 2017. Dynamic relationship between financial development, energy consumption, trade and greenhouse gas: comparison of upper middle income countries from Asia, Europe, Africa and America. *J. Clean. Prod.* 161, 567–580.
- Khurshid, A., Khan, K., Rauf, A., et al., 2024. Effect of geopolitical risk on resources prices in the global and Russian-Ukrainian context: a novel Bayesian structural model. *Res. Policy* 88, 104536.
- Kivimaa, P., Sivonen, M.H., 2023. How will renewables expansion and hydrocarbon decline impact security? Analysis from a socio-technical transitions perspective. *Environ. Innov. Soc. Trans.* 48, 100744.
- Labelle, M.C., 2024. Breaking the era of energy interdependence in Europe: a multidimensional reframing of energy security, sovereignty, and solidarity. *Energy Strat. Rev.* 52, 101314.
- Lee, C.-C., Yuan, Z., He, Z.-W., et al., 2024. Do geopolitical risks always harm energy security? Their non-linear effects and mechanism. *Energy Econ.* 129, 107245.
- Lee, M., Hooy, C.-W., 2012. Country versus industry diversification in ASEAN-5. *Emerg. Mark. Financ. Trade* 49.
- Leifeld, P., Cranmer, S.J., Desmarais, B.A., 2018. Temporal exponential random graph models with btergm: estimation and bootstrap confidence intervals. *J. Stat. Softw.* 83.
- Li, B., Haneklaus, N., 2022. Reducing CO2 emissions in G7 countries: the role of clean energy consumption, trade openness and urbanization. *Energy Rep.* 8, 704–713.
- Li, Y., Liu, Y., Pu, Y., 2024. Dynamic changes in the bauxite trade competition network structure and its influencing factors: based on temporal exponential random graph model. *Res. Policy* 95, 105176.
- Liu, Y., Han, L., Xu, Y., 2021a. The impact of geopolitical uncertainty on energy volatility. *Int. Rev. Financ. Anal.* 75, 101743.
- Liu, Y., Yu, L., Yang, C., et al., 2021b. Heterogeneity of the impact of geopolitical events on energy trade: an empirical study based on regression discontinuity design. *Front. Environ. Sci.* 9.
- Lu, C., Li, B., 2023. The influence factors of innovation networking formation based on ERGM: evidence from the smart medical industry. *J. Digital Econ.* 2, 64–80.
- Ma, X., Yu, T., Jiang, Q., 2023. Does geopolitical risk matter in carbon and crude oil markets from a multi-timescale perspective? *J. Environ. Manag.* 346, 119021.
- Maghyereh, A.I., Awartani, B., Abdoh, H., 2019. The co-movement between oil and clean energy stocks: a wavelet-based analysis of horizon associations. *Energy* 169, 895–913.
- Martínez-García, M.Á., Ramos-Carvajal, C., Cámara, Á., 2023. Consequences of the energy measures derived from the war in Ukraine on the level of prices of EU countries. *Res. Policy* 86, 104114.
- Moiseyev, A., Solberg, B., Kallio, A.M.I., 2014. The impact of subsidies and carbon pricing on the wood biomass use for energy in the EU. *Energy* 76, 161–167.
- Nakhli, S.R., Rafat, M., Dastjerdi, R.B., et al., 2021. Oil sanctions and their transmission channels in the Iranian economy: a DSGE model. *Res. Policy* 70, 101963.
- Ngo, V.M., Nguyen, P.V., Hoang, Y.H., 2024. The impacts of geopolitical risks on gold, oil and financial reserve management. *Res. Policy* 90, 104688.
- Nguyen, H.H., Van Nguyen, P., Ngo, V.M., 2024. Energy security and the shift to renewable resources: the case of Russia-Ukraine war. *Extract. Indust. Soc.* 17, 101442.
- Owjimehr, S., Emami Meybodi, M., Jamshidi, N., 2023. Can geopolitical risk improve energy efficiency in European countries? *Energy Strat. Rev.* 49, 101145.
- Panarello, D., Gatto, A., 2023. Decarbonising Europe – EU citizens perception of renewable energy transition amidst the European Green Deal. *Energy Policy* 172, 113272.
- Qian, L., Zeng, Q., Li, T., 2022. Geopolitical risk and oil price volatility: evidence from Markov-switching model. *Int. Rev. Econ. Financ.* 81, 29–38.
- Qureshi, A., Rizwan, M.S., Ahmad, G., et al., 2022. Russia-Ukraine war and systemic risk: who is taking the heat? *Financ. Res. Lett.* 48, 103036.
- Ren, H., Mu, D., Wang, C., et al., 2024. Vulnerability to geopolitical disruptions of the global electric vehicle lithium-ion battery supply chain network. *Comput. Ind. Eng.* 188, 109919.
- Scholten, D., Bazilian, M., Overland, I., et al., 2020. The geopolitics of renewables: new board, new game. *Energy Policy* 138, 111059.
- Su, C.-W., Khan, K., Umar, M., et al., 2021. Does renewable energy redefine geopolitical risks? *Energy Policy* 158.
- Sweidan, O.D., 2021. Is the geopolitical risk an incentive or obstacle to renewable energy deployment? Evidence from a panel analysis. *Renew. Energy* 178, 377–384.
- Tang, J., Huang, J.-B., Zhang, H.-W., et al., 2021. Time-varying impact of political risk on copper prices. *Trans. Nonferrous Metals Soc. China* 31, 2532–2544.
- Tolochko, P., Boomgaarden, H.G., 2024. Same but different: a comparison of estimation approaches for exponential random graph models for multiple networks. *Soc. Networks* 76, 1–11.
- Vakulchuk, R., Overland, I., Scholten, D., 2020. Renewable energy and geopolitics: a review. *Renew. Sust. Energ. Rev.* 122, 109547.
- Voumik, L.C., Rahman, M.H., Rahman, M.M., et al., 2023. Toward a sustainable future: examining the interconnectedness among foreign direct investment (FDI), urbanization, trade openness, economic growth, and energy usage in Australia. *Reg. Sustainability* 4, 405–415.
- Wang, X., Lee, Z., Wu, S., et al., 2023. Exploring the vital role of geopolitics in the oil market: the case of Russia. *Res. Policy* 85, 103909.
- Wang, X.-Y., Chen, B., Hou, N., et al., 2024. Evolution of structural properties of the global strategic emerging industries' trade network and its determinants: an TERGM analysis. *Ind. Mark. Manag.* 118, 78–92.
- Wang, Y., Liao, M., Wang, Y., et al., 2021. The impact of foreign direct investment on China's carbon emissions through energy intensity and emissions trading system. *Energy Econ.* 97, 105212.
- Wei, N., Xie, W.-J., Zhou, W.-X., 2022. Robustness of the international oil trade network under targeted attacks to economies. *Energy* 251, 123939.
- Wu, G., Feng, L.Y., Peres, M., et al., 2020. Do self-organization and relational embeddedness influence free trade agreements network formation? Evidence from an exponential random graph model. *J. Int. Trade Econ. Dev.* 29, 995–1017.
- Wu, J., Zhao, R., Sun, J., et al., 2023. Impact of geopolitical risks on oil price fluctuations: based on GARCH-MIDAS model. *Res. Policy* 85, 103982.
- Xi, X., Zhou, J., Gao, X., et al., 2019. Impact of changes in crude oil trade network patterns on national economy. *Energy Econ.* 84, 104490.
- Xin, B., Zhang, M., 2023. Evolutionary game on international energy trade under the Russia-Ukraine conflict. *Energy Econ.* 125, 106827.
- Xu, H., Feng, L., Wu, G., et al., 2021. Evolution of structural properties and its determinants of global waste paper trade network based on temporal exponential random graph models. *Renew. Sust. Energ. Rev.* 149, 111402.
- Yan, C., Ji, Y., Chen, R., 2023. Research on the mechanism of selective industrial policies on enterprises' innovation performance — evidence from China's photovoltaic industry. *Renew. Energy* 215, 118868.
- Yang, S., Zhu, Z., Fu, W., et al., 2024. Tele-connection of embodied carbon emissions from industries in China's trade: a complex network analysis. *J. Environ. Manag.* 366, 121652.
- Zhang, H., Wang, Y., Yang, C., et al., 2021. The impact of country risk on energy trade patterns based on complex network and panel regression analyses. *Energy* 222, 119979.
- Zhang, H., Cao, H., Guo, Y., 2024a. The time-varying impact of geopolitical relations on rare earth trade networks: what is the role of China's rare earth export restrictions? *Technol. Forecast. Soc. Chang.* 206, 123550.
- Zhang, L., Wen, S., 2024. The influence of global gallium trade network dynamics on price: a complex network and panel regression approach. *Res. Policy* 93, 105091.
- Zhang, Q., Mao, X., Lu, J., et al., 2024b. EU-Russia energy decoupling in combination with the updated NDCs impacts on global fossil energy trade and carbon emissions. *Appl. Energy* 356, 122415.
- Zhang, S., Shinwari, R., Zhao, S., et al., 2023. Energy transition, geopolitical risk, and natural resources extraction: a novel perspective of energy transition and resources extraction. *Res. Policy* 83, 103608.
- Zheng, S., Zhou, X., Tan, Z., et al., 2022. Preliminary study on the global impact of sanctions on fossil energy trade: based on complex network theory. *Energy Sustain. Dev.* 71, 517–531.
- Zhou, Y., Chen, S., Chen, M., 2019. Global value chain, regional trade networks and Sino-EU FTA. *Struct. Chang. Econ. Dyn.* 50, 26–38.
- Zhu, N., Wang, Y., Yang, S., et al., 2024. Structure characteristics and formation mechanism of the RCEP manufacturing trade network: An ERGM analysis. *Phys. A: Statist. Mech. Appl.* 635, 129488.
- Zhu, X., He, C., Gu, Z., 2021. Benefit from local or destination? The export expansion of the Chinese photovoltaic industry under trade protection. *Energy Econ.* 104, 105690.